

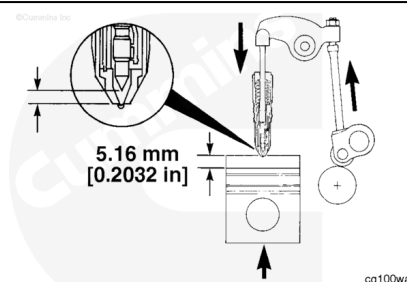
Trainings ▾

General Information

with Mechanically Actuated Injector

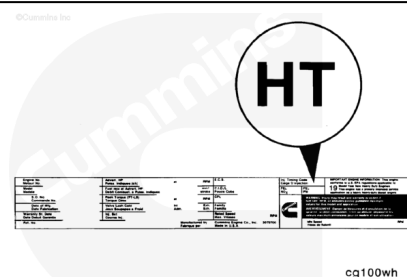
The injection timing is the relative measurement of the distance remaining between the injector plunger and the injector cup when the piston is 5.16 mm [0.2032 in] or 19-degrees before top dead center on the compression stroke.

Injector timing is expressed by the amount of push rod travel remaining.



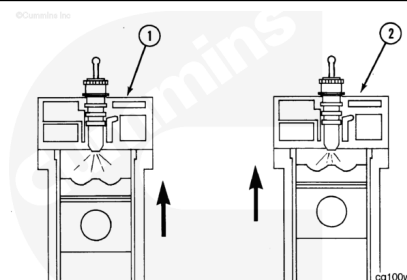
The injection timing code appears on the engine dataplate. Codes are alphabetic letters that relate to a numerical specification.

Specifications can be found in the Control Part List (CPL) Manual, Bulletin 4021328.



The next four frames are a brief review of injection timing and how it can be adjusted.

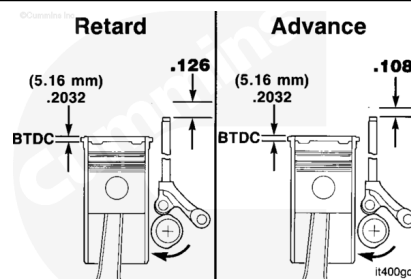
Advanced timing (1) means the fuel is injected earlier into the cylinder during the compression stroke. Retarded timing (2) means the fuel injection occurs in the cylinder closer to top dead center during the compression stroke.



The amount of pushrod travel determines the time of fuel injection in relation to the piston position.

A low numerical value of the pushrod travel remaining indicates a greater degree of advanced (1) or fast timing.

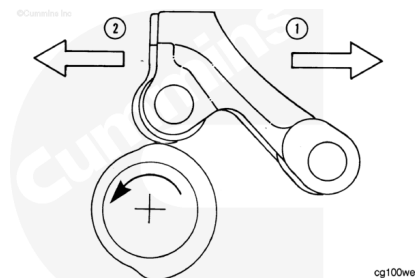
A high numerical value of pushrod travel remaining indicates a greater degree of retarded (2) or slow timing.



Injection timing changes are accomplished by advancing or retarding the cam follower action in relation to the piston position.

This is accomplished by changing the orientation of the camshaft lobe to the cam follower using different camshaft gear keys.

Gear train timing (index mark alignment) **always** remains the same.

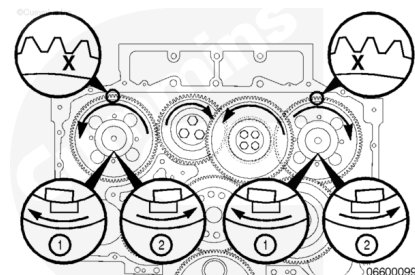


The direction of normal rotation on a QSK45 and QSK60 engine crankshaft is **clockwise** as viewed from the front.

The camshaft key provides a means of indexing the camshaft with the gear.

Offset keys allow the camshaft profile to be rotated slightly while the gear train timing remains the same.

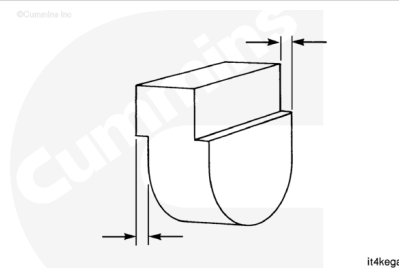
The more the top of the offset is moved in the direction of the camshaft normal rotation, the more the injection timing will be retarded. The pushrod travel numerical value will increase.



If checking or setting the injection timing, it is recommended to use a testing gear. A testing gear is a camshaft gear that has been modified to provide a slip-fit on the camshaft.

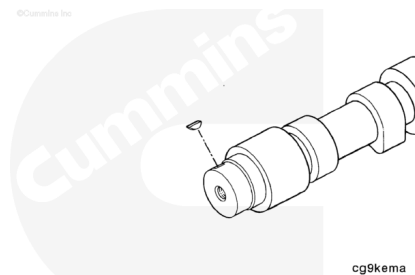
Offset keys can be identified by measuring the offset and referring to the chart at the end of this section.

Each 0.025 mm [0.001 in] of offset will cause a 0.0127 mm [0.0005 in] change in the pushrod travel from a straight key.



with Electronically Actuated Injector

No injection timing adjustment is necessary as it is controlled electronically by the engine control module (ECM). Because of this, **only** a straight key (no offset) **must** be used in the camshaft gear for these engines.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



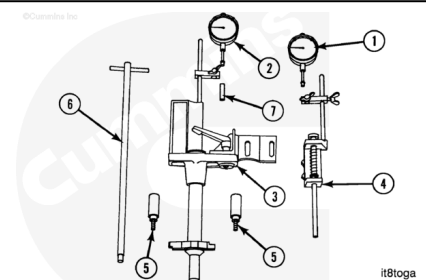
- Disconnect the batteries. See equipment manufacturer service information.
- Remove the rocker lever cover from cylinder number 1 on the right bank. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>)
- Remove the rocker lever assembly from cylinder number 1 on the right bank. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html>)
- Remove the injector from cylinder number 1 on the right bank. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html>)

It is **not** necessary to remove all injectors; however, engine rotation will be easier if all of the injectors are removed.

Setup

Use the injection timing tool, Part Number 3824942, or equivalent. The indicators (1 and 2) are identical.

1. Push rod travel indicator
2. Piston travel indicator
3. Piston plunger support assembly
4. Push rod plunger support assembly
5. Hold-down adapter
6. Extension assembly (adapter wrench)
7. Indicator stem extension.



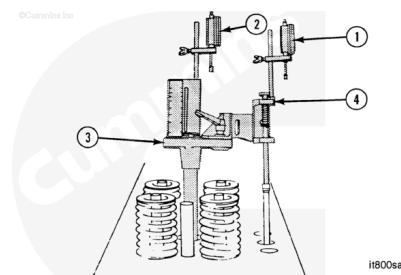
The pushrod plunger support assembly alignment is critical.

Install the pushrod plunger support (4) in the outside slot of the piston plunger support (3).

Align the pushrod plunger support with the mark. Tighten the capscrew.

Install the indicators (1 and 2) on the posts. Turn the indicators so they are **not** over the plungers.

Install the stem extension on the piston travel indicator.

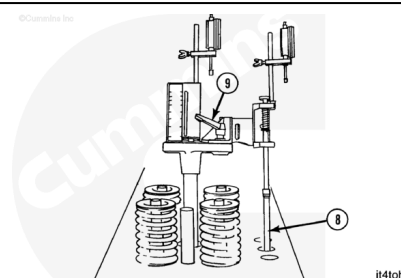


Install the injector pushrod (8) for cylinder number 1 on the right bank.

Install the timing tool in the injector bore of cylinder number 1 on the right bank. Install the hold-down adapters.

Make sure the pushrod plunger and the pushrod are aligned.

Tighten the support lock (9).



Measure

⚠ CAUTION ⚠

Only use the crankshaft to rotate the engine. If any other gear is used, a false measurement will result. Gear lash must be closed up in the direction of normal rotation.

For detailed information on the timing marks. Refer to Procedure 001-008 in Section 1.

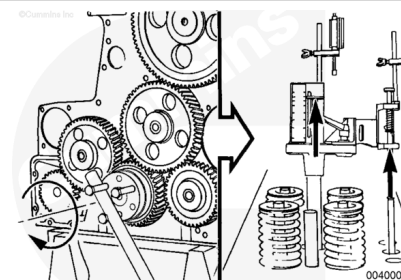
(/qs3/pubsys2/xml/en/procedures/56/56-001-008-tr.html)

Three guide bolts equally spaced in front of the crankshaft will help rotate the engine.

Make sure all of the gear index marks are aligned.

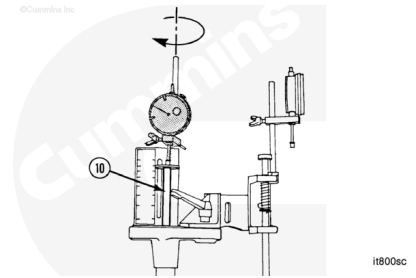
Turn the crankshaft in the direction of normal rotation while observing both of the timing tool plungers. Both plungers will begin moving upward when the cylinder is on the compression stroke.

Assuming all the gear index marks were aligned when the injection timing process was started, the crankshaft will have to be rotated approximately $\frac{3}{4}$ of a revolution to get to the compression stroke for cylinder number 1 on the right bank.



If both plungers are **not** moving up (one up and one down), the engine is on the exhaust stroke. Rotate the crankshaft one revolution to get on the compression stroke.

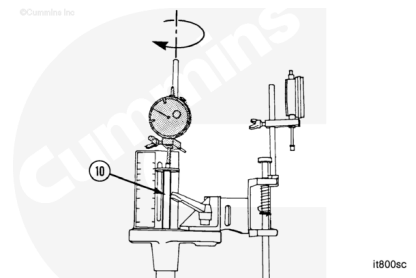
Establish top dead center by slowly rotating the crankshaft in the direction of normal rotation while observing the piston plunger (10). The plunger will move upward, stop, then begin to move downward. The stop point of the plunger is top dead center. Rotate the engine opposite the direction of normal rotation until the plunger begins to move down. The cylinder is now slightly before top dead center.



Turn the indicator so that the stem is touching the plunger. Carefully move the indicator downward until the needle has turned a minimum of five revolutions 12.7 mm [0.500 in]. Lock the indicator in position.

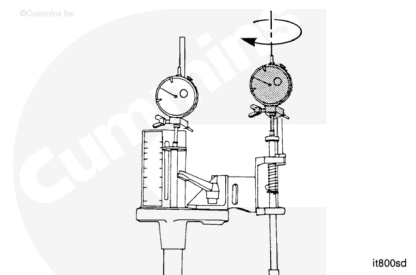
Slowly turn the crankshaft in the direction of normal rotation until the indicator needle stops turning **clockwise** (top dead center). Move the indicator down until there is **only** one revolution, 2.54 mm [0.100 in], of travel remaining until the indicator bottoms out.

Adjust the indicator to zero.



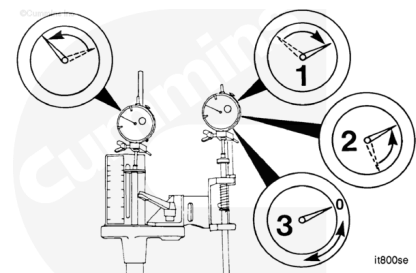
Turn the pushrod indicator so that the stem touches the plunger.

Carefully lower the indicator until it bottoms out. Raise the indicator until the needle has turned a minimum of three revolutions, 7.62 mm [0.300 in].



Slowly turn the crankshaft in the direction of normal rotation until the pushrod indicator stops (1), momentarily reverses direction (2) (this is the crush nose on the camshaft), and stops again (3). The cam follower is now on the outer base circle of the camshaft. The piston is now approximately 45 degrees after top dead center.

It is important to record the amount of travel remaining in the pushrod travel indicator for later reference.



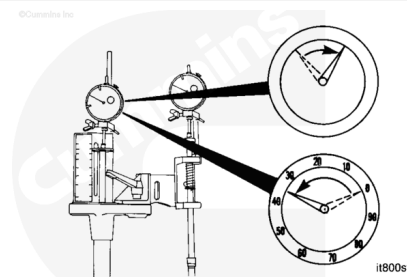
Carefully lower the push rod travel indicator until it bottoms out. Raise the indicator approximately $\frac{1}{2}$ of a revolution 1.27 mm [0.050 in]. Lock the indicator in position.

Set the indicator at zero.

Set the piston at 5.16 mm [0.203 in] before top dead center.

Observe the piston travel indicator as the crankshaft is slowly rotated in the opposite direction of normal rotation.

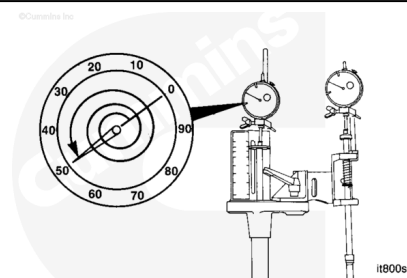
Stop rotating the crankshaft when the piston travel indicator indicates the piston is at top dead center (zero). Recheck top dead center (zero)



The crankshaft **must** be turned slowly to count the indicator revolutions accurately.

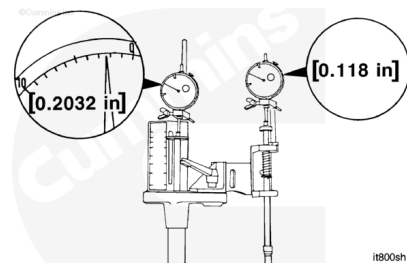
Turn the crankshaft opposite the direction of normal rotation until the indicator needle moves 2- $\frac{1}{2}$ revolutions or 6.35 mm [0.250 in].

The piston is now 6.35 mm [0.250 in] before top dead center.

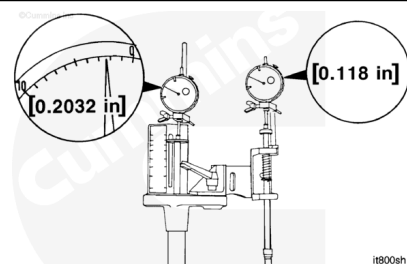


Only move the piston to 5.161 mm [0.203 in] before top dead center by turning the crankshaft in the direction of normal rotation. If the crankshaft is turned too far, it **must** be turned the opposite direction of normal rotation more than 5.161 mm [0.203 in] before top dead center. Then, very slowly turn the crankshaft in the direction of normal rotation until the indicator indicates that the piston is 5.161 mm [0.203 in] before top dead center.

All QSK45 and QSK60 injection timing specifications are more than one indicator revolution, 2.54 mm [0.100 in].



Read the pushrod travel indicator **counterclockwise** from zero. This is the injection timing measurement to compare to the specification. An example of 3 mm [0.118 in] is shown.



A diagram of a microwave oven. Two callout circles point to the interior. The top circle contains a thermometer icon with the text "235°C" and "450°F". The bottom circle contains a clock icon with the text "1-6 hrs". The microwave is shown with its door open, revealing a turntable and a wire rack.

To avoid engine damage, the timing measurement must be confirmed after changing the key.

If the timing is still **not** within specification, the camshaft key **must** be changed. For instructions to remove the camshaft gear. Refer to Procedure 001-012 in Section 1.
(/qs3/pubsys2/xml/en/procedures/56/56-001-012.html)

Repeat this procedure on a front and rear cylinder of the bank(s) from which the camshaft or camshaft key was replaced.

S-302	200704	200706	200708	200709	200711	200714	216294	216782	3000491	3000492	3000493	3000494	3000495	Part Number	Direction of Offset
0.0256	0.0351	0.0411	0.0366	0.0326	0.0289	0.0441	0.0273	0.0311	0.0391	0.0436	0.0471	0.0491	0.0512	3000495	Opposite

Table C															
0.2 35	0.0 33	0.0 39	0.0 34 5	0.0 30 5	0.0 26 8	0.0 42	0.0 25 2	0.0 29	0.0 37	0.0 41 5	0.0 45	0.0 47 1	0.0 49 1	30 00 49 4	Op po sit e
0.2 15	0.0 31	0.0 37	0.0 32 5	0.0 28 5	0.0 24 8	0.0 40	0.0 23 2	0.0 27	0.0 35	0.0 39 5	0.0 43	0.0 44 1	0.0 47 1	30 00 49 3	Op po sit e
0.0 18 5	0.0 28	0.0 34	0.0 29 5	0.0 25 5	0.0 21 8	0.0 37	0.0 20 2	0.0 24	0.0 32	0.0 36 5	0.0 40	0.0 42	0.0 44 1	20 07 14	Op po sit e
0.0 18	0.0 27 5	0.0 33 5	0.0 29	0.0 25	0.0 21 3	0.0 36 5	0.0 19 7	0.0 23 5	0.0 31 5	0.0 36	0.0 39 5	0.0 41 5	0.0 43 6	30 00 49 2	Op po sit e
0.0 15 5	0.0 25	0.0 31	0.0 26 5	0.0 22 5	0.0 18 8	0.0 34	0.0 17 2	0.0 21	0.0 29	0.0 33 5	0.0 37	0.0 39	0.0 41 1	20 07 06	Op po sit e
0.0 13 5	0.0 23	0.0 29	0.0 24 5	0.0 20 5	0.0 16 8	0.0 32	0.0 15 2	0.0 19	0.0 27	0.0 31 5	0.0 35	0.0 37	0.0 39 1	30 00 49 1	Op po sit e
0.0 11	0.0 20 5	0.0 26 5	0.0 22	0.0 18	0.0 14 3	0.0 29 5	0.0 12 7	0.0 16 5	0.0 24 5	0.0 29	0.0 32 5	0.0 34 5	0.0 06 6	20 07 08	Op po sit e
0.0 09 5	0.0 19	0.0 25	0.0 20 5	0.0 16 5	0.0 12 8	0.0 28	0.0 11 2	0.0 15	0.0 23	0.0 27 5	0.0 31	0.0 33	0.0 35 1	20 07 04	Op po sit e
0.0 07	0.0 16 5	0.0 22 5	0.0 18	0.0 14	0.0 10 3	0.0 25 5	0.0 08 7	0.0 12 5	0.0 20 5	0.0 25	0.0 28 5	0.0 30 5	0.0 32 6	20 07 09	Op po sit e
0.0 05 5	0.0 15	0.0 21	0.0 16 5	0.0 12 5	0.0 09	0.0 24	0.0 07 2	0.0 11	0.0 19	0.0 23 5	0.0 27	0.0 29	0.0 31 1	21 67 82	Op po sit e

Table C															
0.0 03 3	0.0 12 8	0.0 18 8	0.0 14 3	0.0 10 3	0.0 06	0.0 21 8	0.0 05	0.0 09	0.0 16 8	0.0 21 3	0.0 24 8	0.0 26 8	0.0 28 9	20 07 11	Op po sit e
0.0 01 7	0.0 11 2	0.0 17 2	0.0 12 7	0.0 08 7	0.0 05 1	0.0 20 2	0.0 03 4	0.0 02	0.0 15 2	0.0 19 7	0.0 23 2	0.0 25 2	0.0 27 3	21 62 94	Op po sit e
0.0 00	0.0 09 5	0.0 15 5	0.0 11	0.0 07	0.0 03	0.0 18 5	0.0 01 7	0.0 06	0.0 13 6	0.0 18	0.0 20 5	0.0 23 5	0.0 25 6	S- 30 2	N/ A
0.0 01 7	0.0 11 2	0.0 17 2	0.0 09 3	0.0 05 3	0.0 01 6	0.0 16 8	0.0 00	0.0 03 8	0.0 11 8	0.0 16 3	0.0 19 8	0.0 21 8	0.0 23 9	21 62 94	Sa me
0.0 03 3	0.0 06 2	0.0 12 2	0.0 07 7	0.0 03 7	0.0 00	0.0 01 52	0.0 01 6	0.0 02	0.0 10 2	0.0 14 7	0.0 18 2	0.0 20 2	0.0 22 3	20 07 11	Sa me
0.0 05 5	0.0 04	0.0 10	0.0 05 5	0.0 01 5	0.0 02	0.0 13	0.0 03 8	0.0 00	0.0 08	0.0 12 5	0.0 16	0.0 18	0.0 20 1	21 67 82	Sa me
0.0 07	0.0 02 5	0.0 08 5	0.0 04	0.0 00	0.0 03 7	0.0 11 5	0.0 05 3	0.0 01 5	0.0 06 5	0.0 11	0.0 14 5	0.0 16 5	0.0 18 6	20 07 09	Sa me
0.0 09 5	0.0 00	0.0 06	0.0 01 5	0.0 02 5	0.0 06 2	0.0 09	0.0 07 8	0.0 04	0.0 04	0.0 08 5	0.0 12	0.0 14	0.0 16 1	20 07 04	Sa me
0.0 11	0.0 01 5	0.0 04 5	0.0 00	0.0 04	0.0 07 7	0.0 07 5	0.0 09 3	0.0 05 5	0.0 02 5	0.0 07	0.0 10 5	0.0 12 5	0.0 14 6	20 07 08	Sa me
0.0 09 5	0.0 00	0.0 06	0.0 01 5	0.0 02 5	0.0 06 2	0.0 09	0.0 07 8	0.0 04	0.0 04	0.0 08 5	0.0 12	0.0 14	0.0 16 1	20 07 04	Sa me
0.0 11	0.0 01 5	0.0 04 5	0.0 00	0.0 04	0.0 07 7	0.0 07 5	0.0 09 3	0.0 05 5	0.0 02 5	0.0 07	0.0 10 5	0.0 12 5	0.0 14 6	20 07 08	Sa me
0.0 13 5	0.0 04	0.0 02	0.0 02 5	0.0 06 5	0.0 10 2	0.0 05	0.0 11 8	0.0 08	0.0 00	0.0 04 5	0.0 08	0.0 10	0.0 12 1	30 00 49 1	Sa me

Table C															
0.0 15 5	0.0 06	0.0 0	0.0 04 5	0.0 08 5	0.0 12	0.0 03	0.0 13 8	0.0 10	0.0 02	0.0 02 5	0.0 06	0.0 08	0.0 10 1	20 07 06	Same
0.0 18	0.0 08 5	0.0 02 5	0.0 07	0.0 11	0.0 14 7	0.0 00 5	0.0 16 3	0.0 12 5	0.0 04 5	0.0 0	0.0 03 5	0.0 05 5	0.0 07 6	30 00 49 2	Same
0.0 18 5	0.0 09	0.0 03	0.0 07 5	0.0 11 5	0.0 15 2	0.0 0	0.0 16 8	0.0 13	0.0 05	0.0 00 5	0.0 03	0.0 05	0.0 07 1	20 07 14	Same
0.0 21 5	0.0 12	0.0 06	0.1 05	0.0 14 5	0.0 18 2	0.0 03	0.0 19 8	0.0 16	0.0 08	0.0 03 5	0.0 00	0.0 02	0.0 04 1	30 00 49 3	Same
0.0 23 5	0.0 14	0.0 08	0.0 12 5	0.0 16 5	0.2 02	0.0 05	0.0 21 8	0.0 18	0.0 10	0.0 05 5	0.0 02	0.0 00	0.0 02 1	30 00 49 4	Same
0.0 25 6	0.0 16 1	0.0 10	0.0 14 6	0.0 18 6	0.0 22 3	0.0 07 1	0.0 23 9	0.0 02 0	0.0 12 1	0.0 07 6	0.0 04 1	0.0 02 1	0.0 00	30 00 49 5	Same

Answer each of the following questions in the spaces provided. The answers to the questions and the use of Tables A, B, and C will determine the timing key required to correct the injection timing.

A working example is attached for your review to illustrate the use of this worksheet.

1.	What is the current timing?	_____
2.	What is the timing code?	_____
3.	What is the timing specification for this code ($\pm$ 0.002 in)?	_____
4.	Is the current timing a larger or smaller number than the specification? - If larger, advance the timing. - If smaller, retard the timing.	_____ _____ _____

5.	What is the difference between the current timing (answer to question 1) and the specification (answer to question 3)?	_____
6.	Does the offset of the current key point in the same or opposite direction that the camshaft normally rotates?	_____
7.	Use Table A to determine the current key part number. - What is the amount of the offset of the current key? - What is the part number of the current key?	_____ _____ _____

Table A

Timing Key Part Number	Offset	
	mm	in
216297	0.09	0.004
200711	0.18	0.007
3826781	0.23	0.009
216782	0.28	0.011
3826782	0.31	0.012
200709	0.38	0.015
3826783	0.43	0.017
200704	0.51	0.020
3826784	0.55	0.022
200708	0.54	0.021
3000491	0.69	0.027
200708	0.84	0.033
3000491	0.91	0.038
200714	0.99	0.039
3000493	1.09	0.043
3000494	1.10	0.47
3000495	1.30	0.051

8.	Use Table B to determine how to use Table C. Circle or check the appropriate answer.	
----	--	--

Table B

Answer to Question 4	Answer to Question 6	Beginning Point on Table C	
Advance	Same	Top of column	_____
Advance	Opposite	Bottom of column	_____
Retard	Same	Bottom of column	_____
Retard	Opposite	Top of column	_____

9.	Answer the following questions BEFORE using Table C to determine the new timing key part number. - Find the current key part number listed at the top of the column on Table C. - Move up or down the column (the answer to question 8). Do not pass 0.000 (zero).	
----	---	--

If you pass 0.000 (zero), you will be choosing a key that does the opposite of what you want it to do.

Stop when you locate the number nearest (± 0.002 in) to the required change in push rod travel (answer to question 5). Remain in this row. Move your finger to the right. The result is the new key part number and direction of offset the timing key **must** point.

- What is the part number of the new key?

Note : Each column on Table C indicates the change in the push rod travel. The change will result if the key at the top of the column is removed and the new key indicated in the second column from the right is installed. The direction of offset will be pointing as indicated in the last column to the right on Table C.

Table C

														Part Number	Direction of Offset
S-302	200704	200706	200708	200709	200711	200714	216294	216782	3000491	3000492	3000493	3000494	3000495		

Table C															
0.0 25 6	0.0 35 1	0.0 41 1	0.0 36 6	0.0 32 6	0.0 28 9	0.0 44 1	0.0 27 3	0.0 31 1	0.0 39 1	0.0 43 6	0.0 47 1	0.0 49 1	0.0 51 2	30 00 49 5	Op po sit e
0.2 35	0.0 33	0.0 39	0.0 34 5	0.0 30 5	0.0 26 8	0.0 42	0.0 25 2	0.0 29	0.0 37	0.0 41 5	0.0 45	0.0 47 1	0.0 49 1	30 00 49 4	Op po sit e
0.2 15	0.0 31	0.0 37	0.0 32 5	0.0 28 5	0.0 24 8	0.0 40	0.0 23 2	0.0 27	0.0 35	0.0 39 5	0.0 43	0.0 44 1	0.0 47 1	30 00 49 3	Op po sit e
0.0 18 5	0.0 28	0.0 34	0.0 29 5	0.0 25 5	0.0 21 8	0.0 37	0.0 20 2	0.0 24	0.0 32	0.0 36 5	0.0 40	0.0 42	0.0 44 1	20 07 14	Op po sit e
0.0 18	0.0 27 5	0.0 33 5	0.0 29	0.0 25	0.0 21 3	0.0 36 5	0.0 19 7	0.0 23 5	0.0 31 5	0.0 36	0.0 39 5	0.0 41 5	0.0 43 6	30 00 49 2	Op po sit e
0.0 15 5	0.0 25	0.0 31	0.0 26 5	0.0 22 5	0.0 18 8	0.0 34	0.0 17 2	0.0 21	0.0 29	0.0 33 5	0.0 37	0.0 39	0.0 41 1	20 07 06	Op po sit e
0.0 13 5	0.0 23	0.0 29	0.0 24 5	0.0 20 5	0.0 16 8	0.0 32	0.0 15 2	0.0 19	0.0 27	0.0 31 5	0.0 35	0.0 37	0.0 39 1	30 00 49 1	Op po sit e
0.0 11	0.0 20 5	0.0 26 5	0.0 22	0.0 18	0.0 14 3	0.0 29 5	0.0 12 7	0.0 16 5	0.0 24 5	0.0 29	0.0 32 5	0.0 34 5	0.0 06 6	20 07 08	Op po sit e
0.0 09 5	0.0 19	0.0 25	0.0 20 5	0.0 16 5	0.0 12 8	0.0 28	0.0 11 2	0.0 15	0.0 23	0.0 27 5	0.0 31	0.0 33	0.0 35 1	20 07 04	Op po sit e
0.0 07	0.0 16 5	0.0 22 5	0.0 18	0.0 14	0.0 10 3	0.0 25 5	0.0 08 7	0.0 12 5	0.0 20 5	0.0 25	0.0 28 5	0.0 30 5	0.0 32 6	20 07 09	Op po sit e

Table C															
0.0 05 5	0.0 15	0.0 21	0.0 16 5	0.0 12 5	0.0 09	0.0 24	0.0 07 2	0.0 11	0.0 19	0.0 23 5	0.0 27	0.0 29	0.0 31 1	21 67 82	Op po sit e
0.0 03 3	0.0 12 8	0.0 18 8	0.0 14 3	0.0 10 3	0.0 06	0.0 21 8	0.0 05	0.0 09	0.0 16 8	0.0 21 3	0.0 24 8	0.0 26 8	0.0 28 9	20 07 11	Op po sit e
0.0 01 7	0.0 11 2	0.0 17 2	0.0 12 7	0.0 08 7	0.0 05 1	0.0 20 2	0.0 03 4	0.0 02	0.0 15 2	0.0 19 7	0.0 23 2	0.0 25 2	0.0 27 3	21 62 94	Op po sit e
0.0 00	0.0 09 5	0.0 15 5	0.0 11	0.0 07	0.0 03	0.0 18 5	0.0 01 7	0.0 06	0.0 13 6	0.0 18	0.0 20 5	0.0 23 5	0.0 25 6	S- 30 2	N/ A
0.0 01 7	0.0 11 2	0.0 17 2	0.0 09 3	0.0 05 3	0.0 01 6	0.0 16 8	0.0 00	0.0 03 8	0.0 11 8	0.0 16 3	0.0 19 8	0.0 21 8	0.0 23 9	21 62 94	Sa me
0.0 03 3	0.0 06 2	0.0 12 2	0.0 07 7	0.0 03 7	0.0 00	0.0 01 52	0.0 01 6	0.0 02	0.0 10 2	0.0 14 7	0.0 18 2	0.0 20 2	0.0 22 3	20 07 11	Sa me
0.0 05 5	0.0 04	0.0 10	0.0 05 5	0.0 01 5	0.0 02	0.0 13	0.0 03 8	0.0 00	0.0 08	0.0 12 5	0.0 16	0.0 18	0.0 20 1	21 67 82	Sa me
0.0 07	0.0 02 5	0.0 08 5	0.0 04	0.0 00	0.0 03 7	0.0 11 5	0.0 05 3	0.0 01 5	0.0 06 5	0.0 11	0.0 14 5	0.0 16 5	0.0 18 6	20 07 09	Sa me
0.0 09 5	0.0 00	0.0 06	0.0 01 5	0.0 02 5	0.0 06 2	0.0 09	0.0 07 8	0.0 04	0.0 04	0.0 08 5	0.0 12	0.0 14	0.0 16 1	20 07 04	Sa me
0.0 11	0.0 01 5	0.0 04 5	0.0 00	0.0 04	0.0 07 7	0.0 07 5	0.0 09 3	0.0 05 5	0.0 02 5	0.0 07	0.0 10 5	0.0 12 5	0.0 14 6	20 07 08	Sa me
0.0 09 5	0.0 00	0.0 06	0.0 01 5	0.0 02 5	0.0 06 2	0.0 09	0.0 07 8	0.0 04	0.0 04	0.0 08 5	0.0 12	0.0 14	0.0 16 1	20 07 04	Sa me
0.0 11	0.0 01 5	0.0 04 5	0.0 00	0.0 04	0.0 07 7	0.0 07 5	0.0 09 3	0.0 05 5	0.0 02 5	0.0 07	0.0 10 5	0.0 12 5	0.0 14 6	20 07 08	Sa me

Table C															
0.0 13 5	0.0 04	0.0 02	0.0 02 5	0.0 06 5	0.0 10 2	0.0 05	0.0 11 8	0.0 08	0.0 0	0.0 04 5	0.0 08	0.0 10	0.0 12 1	30 00 49 1	Same
0.0 15 5	0.0 06	0.0 0	0.0 04 5	0.0 08 5	0.01 2	0.0 03	0.0 13 8	0.0 10	0.0 02	0.0 02 5	0.0 06	0.0 08	0.0 10 1	20 07 06	Same
0.0 18	0.0 08 5	0.0 02 5	0.0 07	0.0 11	0.0 14 7	0.0 00 5	0.0 16 3	0.0 12 5	0.0 04 5	0.0 0	0.0 03 5	0.0 05 5	0.0 07 6	30 00 49 2	Same
0.0 18 5	0.0 09	0.0 03	0.0 07 5	0.0 11 5	0.0 15 2	0.0 0	0.0 16 8	0.0 13	0.0 05	0.0 00 5	0.0 03	0.0 05	0.0 07 1	20 07 14	Same
0.0 21 5	0.0 12	0.0 06	0.1 05	0.0 14 5	0.0 18 2	0.0 03	0.0 19 8	0.0 16	0.0 08	0.0 03 5	0.0 00	0.0 02	0.0 04 1	30 00 49 3	Same
0.0 23 5	0.0 14	0.0 08	0.0 12 5	0.0 16 5	0.2 02	0.0 05	0.0 21 8	0.0 18	0.0 10	0.0 05 5	0.0 02	0.0 00	0.0 02 1	30 00 49 4	Same
0.0 25 6	0.0 16 1	0.0 10	0.0 14 6	0.0 18 6	0.0 22 3	0.0 07 1	0.0 23 9	0.0 02 0	0.0 12 1	0.0 07 6	0.0 04 1	0.0 02 1	0.0 00	30 00 49 5	Same

Answer each of the following questions in the spaces provided. The answers to the questions and the use of Tables A, B, and C will determine the timing key required to correct the injection timing.

A working example is attached for your review to illustrate the use of this worksheet.

1.	What is the current timing?	0.267_____
2.	What is the timing code?	JF_____
3.	What is the timing specification for this code ($\pm$ 0.002 in)?	0.279_____

4.	Is the current timing a larger or smaller number than the specification? • If larger, advance the timing. • If smaller, retard the timing.	Smaller_____ Retard_____
5.	What is the difference between the current timing (answer to question 1) and the specification (answer to question 3)?	0.012_____
6.	Does the offset of the current key point in the same or opposite direction that the camshaft normally rotates?	Same_____
7.	Use Table A to determine the current key part number. • What is the amount of the offset of the current key? • What is the part number of the current key?	0.007_____ 200711_____

Table A

Timing Key Part Number	Offset	
	mm	in
216297	0.09	0.004
200711	0.18	0.007
3826781	0.23	0.009
216782	0.28	0.011
3826782	0.31	0.012
200709	0.38	0.015
3826783	0.43	0.017
200704	0.51	0.020
3826784	0.55	0.022
200708	0.54	0.021
3000491	0.69	0.027
200708	0.84	0.033
3000491	0.91	0.038

Table A		
Timing Key Part Number	Offset	
	mm	in
200714	0.99	0.039
3000493	1.09	0.043
3000494	1.10	0.47
3000495	1.30	0.051
8.	Use Table B to determine how to use Table C. Circle or check the appropriate answer.	

Table B			
Answer to Question 4	Answer to Question 6	Beginning Point on Table C	
Advance	Same	Top of column	_____
Advance	Opposite	Bottom of column	_____
Retard	Same	Bottom of column	_____X_____
Retard	Opposite	Top of column	_____
9.	Answer the following questions BEFORE using Table C to determine the new timing key part number. - Find the current key part number listed at the top of the column on Table C. - Move up or down the column (the answer to question 8). Do not pass 0.000 (zero).		

If you pass 0.000 (zero), you will be choosing a key that does the opposite of what you want it to do.

Stop when you locate the number nearest (± 0.002 in) to the required change in push rod travel (answer to question 5). Remain in this row. Move your finger to the right. The result is the new key part number and direction of offset the timing key **must** point.

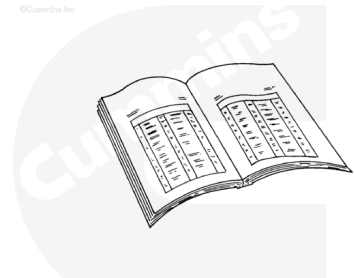
- What is the part number of the new key? **200706**

Note : Each column on Table C indicates the change in the push rod travel. The change will result if the key at the top of the column is removed and the new key indicated in the second column from the right is installed. The direction of offset will be pointing as indicated in the last column to the right on Table C.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

- Install the injector from cylinder number 1 on the right bank. Refer to Procedure 006-026 in Section 6. (</qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html>)
- Install the rocker lever assembly from cylinder number 1 on the right bank. Refer to Procedure 003-009 in Section 3. (</qs3/pubsys2/xml/en/procedures/56/56-003-009-tr.html>)
- Install the rocker lever cover from cylinder number 1 on the right bank. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/56/56-003-011-tr.html>)
- Connect the batteries. See equipment manufacturer service information.